```
postgres=# CREATE DATABASE lab_7;
CREATE DATABASE
postgres=# \c lab_7
You are now connected to database "lab_7" as user "postgres".
lab_7=# CREATE TABLE Departments(
lab_7(# dept_id INTEGER PRIMARY KEY,
lab_7(# dept_name VARCHAR(10) NOT NULL,
lab_7(# location VARCHAR(20) NOT NULL
lab_7(# );
CREATE TABLE
lab_7=# CREATE TABLE Students(
lab_7(# student_id INTEGER PRIMARY KEY,
lab_7(# student_name VARCHAR(20) NOT NULL,
lab_7(# year INTEGER NOT NULL,
lab_7(# dept_id INTEGER,
lab_7(# cgpa DECIMAL NOT NULL,
lab_7(# FOREIGN KEY (dept_id)
lab_7(# REFERENCES Departments (dept_id)
lab_7(# );
CREATE TABLE
lab_7=# CREATE TABLE Courses(
lab_7(# course_id INTEGER PRIMARY KEY,
lab_7(# course_name VARCHAR(30) NOT NULL,
lab_7(# credits INTEGER NOT NULL,
lab_7(# dept_id INTEGER,
lab_7(# FOREIGN KEY(dept_id) REFERENCES Departments(dept_id)
lab_7(# );
CREATE TABLE
lab_7=# CREATE TABLE Enrollments(
lab_7(# enrollment_id INTEGER PRIMARY KEY,
lab_7(# student_id INTEGER,
lab_7(# course_id INTEGER,
lab_7(# marks INTEGER NOT NULL,
lab_7(# FOREIGN KEY(student_id) REFERENCES Students(student_id),
lab_7(# FOREIGN KEY(course_id) REFERENCES Courses(course_id)
lab_7(# );
CREATE TABLE
lab_7=# \d
           List of relations
 Schema |    Name     | Type  |  Owner
--------+-------------+-------+----------
 public | courses     | table | postgres
 public | departments | table | postgres
 public | enrollments | table | postgres
 public | students    | table | postgres
(4 rows)

lab_7=#
```

```
lab_7=# INSERT INTO Departments VALUES
lab_7-# (10, 'CSE', 'Block A'),
lab_7-# (20, 'ECE', 'Block B'),
lab_7-# (30, 'ME', 'Block C'),
lab_7-# (40, 'Design', 'Block D');
INSERT 0 4
lab_7=# SELECT * FROM Departments;
 dept_id | dept_name | location
---------+-----------+----------
      10 | CSE       | Block A
      20 | ECE       | Block B
      30 | ME        | Block C
      40 | Design    | Block D
(4 rows)

lab_7=# INSERT INTO Students VALUES
lab_7-# (1001, 'Ananya', 2, 10, 8.60),
lab_7-# (1002, 'Bharath', 3, 20, 7.80),
lab_7-# (1003, 'Charan', 2, 10, 9.10),
lab_7-# (1004, 'Divya', 4, 30, 8.20),
lab_7-# (1005, 'Farhan', 3, 20, 7.40),
lab_7-# (1006, 'Gita', 1, 40, 8.90);
INSERT 0 6
lab_7=# SELECT * FROM Students;
 student_id | student_name | year | dept_id | cgpa
------------+--------------+------+---------+------
       1001 | Ananya       |    2 |      10 | 8.60
       1002 | Bharath      |    3 |      20 | 7.80
       1003 | Charan       |    2 |      10 | 9.10
       1004 | Divya        |    4 |      30 | 8.20
       1005 | Farhan       |    3 |      20 | 7.40
       1006 | Gita         |    1 |      40 | 8.90
(6 rows)
```

```
lab_7=# INSERT INTO Courses VALUES
(501, 'Database Systems', 4, 10),
(502, 'Computer Networks', 3, 10),
(503, 'Digital Systems', 4, 20),
(504, 'Manufacturing', 3, 30),
(505, 'Design Thinking', 2, 40);
INSERT 0 5
lab_7=# SELECT * FROM Courses;
 course_id |    course_name    | credits | dept_id
-----------+-------------------+---------+---------
       501 | Database Systems  |       4 |      10
       502 | Computer Networks |       3 |      10
       503 | Digital Systems   |       4 |      20
       504 | Manufacturing     |       3 |      30
       505 | Design Thinking   |       2 |      40
(5 rows)

lab_7=# INSERT INTO Enrollments VALUES
lab_7-# (9001, 1001, 501, 86),
lab_7-# (9002, 1001, 502, 78),
lab_7-# (9003, 1002, 503, 74),
lab_7-# (9004, 1003, 501, 92),
lab_7-# (9005, 1004, 504, 81),
lab_7-# (9006, 1005, 503, 69);
INSERT 0 6
lab_7=# SELECT * FROM Enrollments;
 enrollment_id | student_id | course_id | marks
---------------+------------+-----------+-------
          9001 |       1001 |       501 |    86
          9002 |       1001 |       502 |    78
          9003 |       1002 |       503 |    74
          9004 |       1003 |       501 |    92
          9005 |       1004 |       504 |    81
          9006 |       1005 |       503 |    69
(6 rows)

lab_7=#
```

```
lab_7=# UPDATE Students SET cgpa = 8.15 WHERE student_id = 1002;
UPDATE 1
lab_7=# SELECT * FROM Students WHERE student_id = 1002;
 student_id | student_name | year | dept_id | cgpa
------------+--------------+------+---------+------
       1002 | Bharath      |    3 |      20 | 8.15
(1 row)

lab_7=# DELETE FROM Enrollments WHERE enrollment_id = 9006;
DELETE 1
lab_7=# SELECT * FROM Enrollments WHERE enrollment_id = 9006;
 enrollment_id | student_id | course_id | marks
---------------+------------+-----------+-------
(0 rows)

lab_7=#
```

```python
import psycopg

connection = psycopg.connect(
        dbname = "lab_7",
        user = "postgres",
        password = "509",
        host = "localhost",
        port = "5432"
)

cursor = connection.cursor()

print("Connected to PostgreSQL")

cursor.execute("""
        CREATE VIEW CSE_Students_View AS
        SELECT student_id, Student_name, year, cgpa
        FROM Students
        WHERE dept_id = 10;
""")

print("\n CSE_STUDENTS_VIEW created.")

cursor.execute("""
        SELECT * FROM CSE_Students_View ORDER BY student_id;
""")

for row in cursor.fetchall():
        print(row)

cursor.execute("""
        UPDATE CSE_Students_View
        SET cgpa = 8.95
        WHERE student_id = 1001;
""")

print("\n Updated cgpa of student with student_id 1001 as 8.95")

print("\nVIEW")

cursor.execute("""
        SELECT * FROM CSE_Students_View ORDER BY student_id;
""")

for row in cursor.fetchall():
        print(row)

print("\nTable")

cursor.execute("""
        SELECT student_id, student_name, year, cgpa FROM Students;
""")

for row in cursor.fetchall():
```

1,1 Top

```
cursor.execute("""
        SELECT student_id, student_name, year, cgpa FROM Students;
""")

for row in cursor.fetchall():
        print(row)

cursor.execute("""
        CREATE VIEW Student_Enrollment_View AS
        SELECT * FROM Enrollments;
""")

print("\nCreated view Student_Enrollment_View")

cursor.execute("""
        SELECT * FROM Student_Enrollment_View;
""")

for row in cursor.fetchall():
        print(row)

cursor.execute("""
        DELETE FROM Student_Enrollment_View
        WHERE enrollment_id = 9002;
""")

print("\nDeleted row with enrollemtn_id = 9002")

cursor.execute("""
        SELECT * FROM Student_Enrollment_View;
""")

for row in cursor.fetchall():
        print(row)

cursor.execute("""
        DROM VIEW CSE_Students_View;
""")

cursor.execute("""
        DROM VIEW Student_Enrollment_View;
""")

print("\nDeleted all views")

connection.commit()

print("\nAll operations completed successfully")

cursor.close()
connection.close()

                                                                    102,1-8       Bot
```

```
L509@L509:~/cs25b1096/DBMS/lab-7$ python3 lab-7.py
Connected to PostgreSQL

 CSE_STUDENTS_VIEW created.
(1001, 'Ananya', 2, Decimal('8.95'))
(1003, 'Charan', 2, Decimal('9.10'))

 Updated cgpa of student with student_id 1001 as 8.95

VIEW
(1001, 'Ananya', 2, Decimal('8.95'))
(1003, 'Charan', 2, Decimal('9.10'))

Table
(1003, 'Charan', 2, Decimal('9.10'))
(1004, 'Divya', 4, Decimal('8.20'))
(1005, 'Farhan', 3, Decimal('7.40'))
(1006, 'Gita', 1, Decimal('8.90'))
(1002, 'Bharath', 3, Decimal('8.15'))
(1001, 'Ananya', 2, Decimal('8.95'))

Created view Student_Enrollment_View
(9001, 1001, 501, 86)
(9003, 1002, 503, 74)
(9004, 1003, 501, 92)
(9005, 1004, 504, 81)

Deleted row with enrollemtn_id = 9002
(9001, 1001, 501, 86)
(9003, 1002, 503, 74)
(9004, 1003, 501, 92)
(9005, 1004, 504, 81)

Deleted all views
```